

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

03

10 answers · Rationale-rich · Teach from doc

Q1**A****A.** 234

Choice A is correct. It's given that a special camera is used for underwater ocean research, and this camera is at a depth of 39 fathoms. It's also given that 1 fathom is equal to 6 feet. Thus, 39 fathoms is equivalent to $39 \text{ fathoms} \frac{6 \text{ feet}}{1 \text{ fathom}}$, or 234 feet. Therefore, the camera's depth, in feet, is 234.

Choice B is incorrect. This is the camera's depth, in feet, if the camera is at a depth of 19.5 fathoms.

Choice C is incorrect. This is the camera's depth, in feet, if the camera is at a depth of 7.5 fathoms.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**C**

C. Plausible values of the percent of the students at the school who support a menu change are between 32.5% and 43.5%.

Choice C is correct. It's given that an estimated 38% of sampled students at the school were in support of a menu change, with a margin of error of 5.5%. It follows that the percent of the students at the school who support a menu change is 38% plus or minus 5.5%. The lower bound of this estimation is $38 - 5.5$, or 32.5%. The upper bound of this estimation is $38 + 5.5$, or 43.5%. Therefore, plausible values of the percent of the students at the school who support a menu change are between 32.5% and 43.5%.

Choice A is incorrect. This is the percent of the sampled students at the school who support a menu change. Choices B and D are incorrect and may result from misinterpreting the margin of error.

Q3**240**

The correct answer is 240. It's given that 80% of the 300 seeds sprouted. Therefore, the number of seeds that sprouted can be calculated by multiplying the number of seeds that were planted by $\frac{80}{100}$, which gives $300\frac{80}{100}$, or 240.

Q4**B****B. 252**

Choice B is correct. The table summarizes all 1,000 responses from the students surveyed. If 312 are males who play a sport, 220 are females who play a sport, and 216 are females who do not play a sport, then $1,000 - 312 - 220 - 216 = 252$ males who do not play a sport.

Choices A, C, and D are incorrect. If 109 males who do not play a sport responded, then the table summary would be $109 + 312 + 220 + 216 = 857$ total student responses rather than 1,000. If 468 males who do not play a sport responded, then the table summary would be $468 + 312 + 220 + 216 = 1,216$ total student responses rather than 1,000. If 688 males who do not play a sport responded, then the table summary would be $688 + 312 + 220 + 216 = 1,436$ total student responses rather than 1,000.

Q5**D**

D. 18

Choice D is correct. The height of each bar in the graph shown represents the number of volunteers who chose the gift labeled with the letter specified at the bottom of the bar. The bar for gift C has a height of 18. Therefore, 18 volunteers chose gift C.

Choice A is incorrect. This is the number of volunteers who chose gift D, not gift C.

Choice B is incorrect. This is the number of volunteers who chose gift B, not gift C.

Choice C is incorrect. This is the number of volunteers who chose gift A, not gift C.

Q6**B**

B. 35

Choice B is correct. In the given scatterplot, the x -values represent the distance above sea level, in feet, and the y -values represent the temperature, in $^{\circ}\text{F}$. The point on the line of best fit with an x -value of 4,000 has a corresponding y -value of 35. Therefore, at a distance of 4,000 feet above sea level, the temperature predicted by the line of best fit is 35°F .

Choice A is incorrect. This is the temperature, in $^{\circ}\text{F}$, predicted by the line of best fit at a distance of 0 feet above sea level.

Choice C is incorrect. This is the measured temperature, in $^{\circ}\text{F}$, at a distance of 6,000 feet above sea level.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A**A. $8u$

Choice A is correct. It's given that $t = 4u$. Multiplying both sides of this equation by 2 yields $2t = 2(4u)$, or $2t = 8u$.

Choice B is incorrect and may result from dividing, instead of multiplying, the right-hand side of the equation by 2. Choices C and D are incorrect and may result from calculation errors.

Q8**B**

B. 2,100

Choice B is correct. Let x be the number of people in the entire town that would be expected to name chocolate. Since the sample of 50 people was selected at random, it is reasonable to expect that the proportion of people who named chocolate as their favorite ice-cream flavor would be the same for both the sample and the town population. Symbolically, this can be expressed as $\frac{7}{50} = \frac{x}{14,878}$. Using cross multiplication, $7 \times 14,878 = x \times 50$; solving for x yields 2,083. The choice closest to the value of 2,083 is choice B, 2,100.

Choices A, C, and D are incorrect and may be the result of errors when setting up the proportion, solving for the unknown, or incorrectly comparing the choices to the number of people expected to name chocolate, 2,083.

Q9**B****B.** 14

Choice B is correct. It's given that 4% of the 350 objects in the box are balls. Therefore, the number of balls in the box can be found by calculating 4% of 350, which is equal to $350 \frac{4}{100}$, or 14.

Choice A is incorrect. This is the percentage of objects in the box that are balls, not the number of balls in the box.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**B****B.** $\frac{1}{3}$

Choice B is correct. If a number from the data set is selected at random, the probability of selecting a negative number is the count of negative numbers in the data set divided by the total count of numbers in the data set. It's given that a data set of three numbers is shown. It follows that the total count of numbers in the data set is 3. In the data set shown, -13 is the only negative number. It follows that the count of negative numbers in the data set is 1. Therefore, if a number from the data set is selected at random, the probability of selecting a negative number is $\frac{1}{3}$.

Choice A is incorrect. This is the probability of selecting a negative number from a data set that doesn't contain any negative numbers.

Choice C is incorrect. This is the probability of selecting a positive number, not a negative number, from the data set.

Choice D is incorrect. This is the probability of selecting a negative number from a data set that contains only negative numbers.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

